

SIGHTLINE — Project Spec

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A presentation evaluation tool that measures two things no existing product measures: where a viewer's eye actually goes, and what different viewers actually take away.

1. Thesis

Presentation feedback tools check delivery and formatting — filler words, pace, eye contact, font size. None check whether the audience extracts the point the presenter meant, and none check whether that extraction *differs depending on who is watching*.

SIGHTLINE simulates three audiences separated by **prior knowledge**, has each one attempt comprehension of a slide, and measures divergence between them. High novice-to-expert divergence localizes the **expert blind spot** to a specific slide — the single most common failure mode in technical presentations, and one the presenter is structurally unable to detect alone.

A saliency layer adds *where the eye goes*. An optional neural layer adds *which cortical systems the slide drives*.

One sentence: We show you the wrong conclusion your audience will draw, and which slide caused it.

2. Prior art — what the research turned up

2.1 The generic version is a crowded commodity

"Upload a talk, get feedback on engagement and clarity" is a mature category. Yoodli tracks 12+ metrics including pacing, filler words, eye contact and clarity. Orai, Poised, Insight7, Yudly, VirtualSpeech and Presentace occupy the same space. Microsoft ships Speaker Coach free inside PowerPoint.

Implication: do not pitch a presentation coach. Pitch the divergence measurement.

2.2 A published null result sits directly on the naive neural approach

arXiv **2607.01400** (July 2026) ran TRIBE on 48 YouTube videos, reduced the predicted cortical response to a scalar engagement curve via global field power, and tested it against "most replayed" heatmaps.

Signal	Pooled partial <i>r</i> (position-controlled)
TRIBE engagement (GFP)	+0.058 (95% CI -0.04 to 0.15, <i>p</i> = 0.23)
Loudness baseline	+0.040
Motion baseline	-0.061

The null held across six cortical-network readouts (visual, auditory, salience, frontal, parietal, whole-cortex) and under a circular-shift permutation test preserving autocorrelation. Video-level ranking also failed: *p* with views and likes near zero and slightly negative.

Implication: a scalar "engagement score" derived from predicted brain drive is a known-false output. It must not appear in the product.

2.3 But the measured-brain literature is solidly positive

- Meta-analysis of 14 studies: intersubject correlation (ISC) correlates with attention at **r = 0.65**.
- Cohen et al. (2018): EEG ISC during educational videos predicted information retention; when attention was divided by a counting task, ISC dropped significantly and stopped predicting exam performance.
- Classroom EEG studies reproduce ISC engagement effects with portable consumer-grade equipment.

The gap between 2.2 and 2.3 is a readout problem. GFP measures *amplitude*; ISC measures *reliability*. Loud, flashy content produces amplitude. Only followable content produces synchrony.

2.4 Additional constraint — the seductive allure effect

Brain images and neuroscience language increase perceived credibility of an explanation independent of whether the explanation is sound. Any cortical visualization we ship will do this to judges. **Naming this ourselves in the methods panel converts it from a liability into a credibility signal.**

3. What we build

Three layers, two speeds

Layer	What it answers	Speed	Demo path
Saliency	Where does the eye land first?	< 1 s, CPU	Live
Simulated audiences	What do three viewers think the point is?	< 5 s	Live
Neural	Which cortical systems does this slide drive?	6–13 min GPU/slide	Precomputed only

The live path is the product. The neural layer is the proof-of-concept centerpiece, run overnight on two bundled sample decks.

Core design rules — non-negotiable

1. **Audiences perform, they never rate.** No **clarity: 7/10** anywhere in the codebase. An audience outputs a takeaway, a confidence, and unresolved terms. Those are checkable; a rating is not.
 2. **Every number decomposes into its source text.** Clicking a score shows the three takeaways that produced it.
 3. **No scalar engagement metric.** See §2.2. If one appears, it is a bug. GFP is reproduced *only* as a labeled negative baseline in the methods panel.
 4. **Neural output is labeled "predicted — simulated, not measured"** on the image itself, not in a tab.
 5. **Ratios within a deck, never absolutes.** Within-deck comparison holds presenter, topic and recording conditions fixed and cancels much of the amplitude confound.
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4. The three audiences

Defined **only by prior knowledge**. Not by personality, demographic or vibe — prior knowledge is the axis with theory behind it.

Persona	Definition
novice	No domain background, general education
peer	Adjacent technical field, not this subfield
expert	Deep in this exact subfield

Each receives the slide image, extracted text, and running deck context. Each returns strict JSON:

```
{
  "takeaway": "one sentence — what they think the point is",
  "confidence": 0.0,
  "unresolved_terms": ["terms used but not defined for THIS audience"],
  "questions": ["what they'd need to ask to follow"],
  "inferred_claim": "what they think the presenter wants them to believe"
}
```

Run concurrently. Cache by (*slide_hash*, *persona*). **Never let a persona see another persona's output.**

Derived metrics

- **intent_alignment** — semantic similarity between each audience's takeaway and the presenter's declared intent. Three values per slide.
- **audience_divergence** — pairwise semantic distance between the three takeaways. **This is the headline metric.**
- **blind_spot_score** — expert alignment minus novice alignment. Large positive = classic expert blind spot.
- **term_gap** — terms in *novice.unresolved_terms* absent from *expert.unresolved_terms*. The specific words costing you the room.

5. Neural layer (precomputed)

What we extract: *where*, not *how much*

The valuable signal from TRIBE is spatial. Collapsing to a scalar throws it away and reproduces a known null.

Question per slide: does it drive language regions, or only early visual?

A slide producing visual-cortex drive with little language drive is one people are *looking at* without *processing meaning from* — decorative, dense, pretty, empty. A slide driving language and association cortex is being read and understood.

processing_ratio = language_drive / visual_drive, expressed as a Z-score relative to the deck mean.

Slide 12 drives language 40% below deck average — a claim with content. Slide 12 scores 61 — not.

This ratio is a **proposed readout, not a validated metric**, and the write-up must say so.

Hard constraints

- **TRIBE requires narrated slides.** On a static image V-JEPA2 contributes nearly nothing — signal comes almost entirely from the audio and transcript encoders. A silent PDF gives TRIBE nothing. Disable the layer entirely and say so rather than fabricating from text alone.
- **6–13 min GPU per slide.** Live inference is impossible. Precompute only, via a standalone resumable CLI.
- **Out of distribution.** TRIBE was trained on movie-watching fMRI. Narrated slides are not movies. State this.

Convergence

The processing ratio and the novice/expert divergence measure related things through **completely independent routes**. Where they agree → converging evidence. Where they disagree → something interesting to say.

6. Validation — the thing that separates us from a demo

A **/validate** route that shows a slide and collects (a) a one-sentence takeaway and (b) a self-reported familiarity rating, appending to **validation/{deck_id}.json**.

On-site protocol: two slides, ~8 hackers who know the topic, ~8 who don't, one sentence each. Compare real novice/expert divergence against simulated divergence.

Twenty minutes of work. It is the difference between *"we made LLM personas"* and *"we made LLM personas and checked whether they behave like real people."*

Recruit during the 2am lull when everyone wants an excuse to stop coding.

7. Build order

Respect this order. Each step is a working checkpoint.

1. **audiences.py + divergence.py with tests**. Test on two hand-written slides where the answer is known: one obviously clear (all three converge), one loaded with undefined jargon (novice *must* diverge). **If those two don't behave correctly, the project is unsound — fail loudly here.**
2. Slide detail screen wired to those modules. PDF ingest only.
3. **saliency.py + scanpath.py** with synthetic-image tests (one white square on black → one fixation at its center).
4. **diagnose.py** rules.
5. Blind spot view.
6. **/validate** route.
7. pptx ingest.
8. **fix.py** — the revise-and-rescore loop.
9. Neural layer, **only if 1–6 are done.**

Cut order under time pressure

Cut from the bottom. **If you must cut, cut the fix loop and the neural layer — never the audiences.** Saliency alone is a nice linter; divergence is a finding.

Hard checkpoint

If step 1's jargon slide does not produce novice/expert divergence by **hour 3**, the personas aren't differentiating. Stop and fix the prompting, or the rest of the build rests on nothing.

8. Technical stack

Python FastAPI + React/Vite. Single repo, **make dev** runs both.

Ingest: **python-pptx** for pptx, **pdfplumber** + **pdf2image** for pdf. Render slides to 1280×720 PNG. Cap 30 slides.

Saliency: **SaliencyModel** protocol with three implementations behind it —

- **SpectralResidual** (OpenCV) — CPU, instant, zero deps. **Default and demo path.**
- **DeepGaze** — DeepGaze IIE if weights present, else clear startup error.
- **Stub** — deterministic, for tests.

Scanpath: greedy max + inhibition-of-return (zero a $\sigma=60\text{px}$ Gaussian), $N=6$ fixations. Document that this *approximates* a scanpath; it is not a validated scanpath model.

Semantic similarity: `sentence-transformers`, local model, no API call.

Neural: `NeuralModel` protocol $\rightarrow (T, 20484)$ on fsaverage5, $TR=1\text{s}$. `CachedNeural` is the only path available to the web app; `TribeNeural` raises immediately if called from the FastAPI process. Parcellation via `nilearn.datasets` (Schaefer or Destrieux). Rendering via `nilearn.plotting.plot_surf_stat_map`, colormap `cold_hot`, symmetric about zero.

Offline: fully offline capable except LLM calls. Bundle cached audience responses for sample decks so the demo survives dead wifi.

9. Screens

1. **Upload** — drag-drop, plus two bundled sample decks (one clean, one with a deliberate expert blind spot), fully cached.
2. **Deck overview** — thumbnail grid, two small bars per slide (attention alignment, audience divergence). Sort worst divergence first.
3. **Slide detail** — split view. Left: slide with saliency heatmap (viridis, adjustable opacity) and numbered fixation path. Right: three audience cards, each showing its takeaway verbatim, confidence, unresolved terms. Declared intent above them for comparison.
4. **Blind spot view** — deck-level chart: novice vs. expert alignment per slide, widest gaps called out. **This is the screen that sells the demo.**
5. **Neural view** — sample decks only. Cortical surface, lateral + medial, both hemispheres. Permanent overlay: *"Predicted response — simulated, not measured."* Processing ratio as deviation from deck mean, deck distribution shown behind it. Clear empty state when no cached data — never a placeholder brain.
6. **Fix view** — before/after on both metrics. If the fix doesn't improve, show the failed attempt and say so.
7. **Methods** — see §10.

Design: editorial, high contrast, restrained. Slide image is the hero. Three audience cards read as three distinct voices — differentiate by typography and a subtle left-border accent, not cartoon avatars. Monospace for all numbers. Viridis or inferno for heatmaps, never rainbow. No gradients, no emoji.

10. Methods panel — required disclosures

Ship all of these. They are what make the project defensible rather than impressive-looking.

- Saliency predicts **bottom-up attention**, not comprehension or aesthetics.
- The scanpath is an inhibition-of-return approximation, not a validated scanpath model.
- Simulated audiences are **LLM personas**, not validated against human readers except where §6 data exists.
- TRIBE was trained on movie-watching fMRI; narrated slides are **out of distribution**.
- Neural inference costs 6–13 min GPU/slide; all neural results shown are **precomputed**.
- The language/visual ratio is a **proposed readout, not a validated metric**.
- **Brain images increase perceived credibility independent of the underlying claim's validity**. State what the image does and does not establish.
- Reproduce the GFP readout as a labeled negative baseline alongside the published null it replicates.

11. Sponsor targeting

Sponsor	Fit	Notes
Long Lake	Strongest	A skeptic shown a wrong conclusion drawn from their own slide is exactly the "one great experience away" brief. Lead here.
Voloridge	Strong	Signal-vs-noise on public data; we separate a real signal from a position artifact. \$5,000 first place.
ASUS	Clean	If the GX10 is used for the precompute run. Claim it in hour one — first come, first serve.
Dropbox	Good	"Turn files into action" — a deck becomes a diagnosis. Fits the brief better than most.
Arrowstreet	Possible	Judged on evidence quality, source citation and conclusion confidence. Our CIs and baselines map well,

Sponsor	Fit	Notes
		but the greenwashing topic is fixed. Only with spare submission capacity.
Token Company	Cheap	Document caching and model-tiering on the persona calls. Three personas × 30 slides is a real cost story. \$500.
Deepgram	Cheap	Transcription for narrated decks.
Meta	Weak	Brief is human connection, not education tooling. Worth noting TRIBE is Meta research (d'Ascoli, King et al.), but that doesn't fix the mismatch.
ElevenLabs	Drop	Their criteria explicitly deprioritize simple TTS. Only viable with an agentic re-narration loop.

12. Demo-night checklist

- ☐ Two sample decks load in < 1 s with fully cached audience responses
- ☐ Works with wifi disabled
- ☐ Neural precompute finished, or the neural view cleanly hidden
- ☐ Blind spot view is the first screen a judge sees after upload
- ☐ Validation overlay populated with real human data
- ☐ "Predicted — simulated, not measured" visible on every cortical image
- ☐ 2-minute pitch rehearsed to muscle memory
- ☐ Demo video recorded as backup before the venue gets loud

13. The pitch

Presentation tools check your delivery. Nobody can tell you whether your *content* landed — or that it landed differently for the person who already knew the topic than for the one who didn't.

SIGHTLINE simulates three audiences separated by prior knowledge, has each one tell us what they think your point was, and shows you where they disagree. That disagreement is your expert blind spot, localized to a single slide.

We tested a brain-encoding model as a fourth signal, found the published scalar readout fails, and report a spatial one instead — labeled honestly as the hypothesis it is.

What to claim: we found the wrong conclusion your audience draws. **What not to claim:** a number that says how engaging your slide is.